

Emotion-Conditioned Speech Synthesis Using a Diffusion Transformer Fine-Tuned on MELD *

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Abstract

Making computers talk with the right emotion is a hard problem. Current text-to-speech (TTS) systems can produce clear speech, but most of them sound flat and emotionless. In this work I try to fix that by fine-tuning a Large Diffusion Transformer TTS model, called **LongCat-AudioDiT-1B**, on the MELD dataset — a dataset of real conversations from the TV series *Friends*, annotated with seven emotion labels. I use a technique called Low-Rank Adaptation (LoRA) so I can update only a small part of the model and still fit it on a consumer-grade GPU. I condition the model on emotion by prepending special emotion tokens such as [JOY] or [ANGER] to the input text, which is then encoded by a frozen T5 encoder. I compare my system against two references: a small manually implemented DiT baseline trained from scratch, and **UMETTS** (Cheng et al., 2024), a strong published baseline for emotional TTS on MELD that achieves a Mel-Cepstral Distortion (MCD) of 5.93 dB. The manual DiT baseline yields a mean MCD of **49.44 dB**. My fine-tuned LongCat-AudioDiT-1B, evaluated at global checkpoint step 1700 (almost one full epoch on MELD), achieves a mean MCD of **18.49 dB** — a reduction of more than 30 dB over the manual baseline. The surprise emotion reaches a particularly low MCD of **6.05 dB**, near the UMETTS score. I discuss what drives these results, where the remaining gap to the published baseline comes from, and how further training and a better vocoder would close it.

1 Introduction

The objective of this work is to develop a system capable of generating speech that is emotionally accurate, moving beyond simple clarity to express specific affects such as anger, happiness, and sadness. The goal is to produce audio that a listener

can readily recognize as carrying a designated emotion given a sentence and a corresponding label. While modern text-to-speech (TTS) systems are highly clear and clean, they often lack emotional depth, sounding more like professional news anchors than conversational human beings. Providing speech that sounds human and emotionally alive is critical for the development of virtual assistants, social robots, audiobooks, and tools for individuals with communication difficulties.

Historically, emotional TTS systems functioned by utilizing emotion labels as auxiliary inputs or by cloning the emotion from a reference audio clip. These approaches are often limited by the high cost and rarity of labeled emotional speech data and frequently fail to generalize beyond a small number of emotion categories. While recent diffusion-based TTS models have achieved greater naturalness, they are typically trained on neutral, clean studio recordings. Consequently, they are often ill-equipped to handle the messy, conversational, and highly emotional speech found in real-world settings. Currently, the leading baseline for emotional TTS on the MELD dataset is UMETTS, which achieves a Mel-Cepstral Distortion (MCD) of 5.93 dB using a complex multimodal alignment module. This study investigates whether a simpler approach—fine-tuning a large pre-trained Diffusion Transformer (DiT) with emotion tokens—can achieve competitive quality.

In this approach, we fine-tune LongCat-AudioDiT-1B, a billion-parameter Diffusion Transformer, on the MELD dataset using Low-Rank Adaptation (LoRA). Conditioning is achieved by prepending special emotion tokens, such as [JOY] or [ANGER], to the text prompt before it is processed by a frozen T5 encoder. Additionally, a smaller Diffusion Transformer was implemented manually from scratch to serve as a baseline for understanding the architecture and training dynamics. Successfully implementing this would make

*Code, models, and evaluation scripts are available at <https://github.com/HellDragger/SenticDiT>.

emotional TTS significantly more accessible, allowing developers to add emotional control with a simple fine-tuning recipe rather than a complex multimodal pipeline. Such accessibility is vital for assistive technology, gaming, and language learning tools.

There are inherent risks in fine-tuning a large diffusion model on a relatively small conversational dataset of approximately 9,000 samples, including the potential for catastrophic forgetting or the failure to learn the emotion signal due to data noise. Preliminary results suggest that achieving high-quality emotional adaptation requires more careful tuning and compute than initially expected. All experiments were performed using free Kaggle T4 GPUs and Google Colab, incurring no extra costs. The manual DiT experiment was conducted over 100 epochs, while the AudioDiT fine-tuning reached one full epoch (approximately 1,900 steps) before compute limits were reached.

The primary metric for evaluation is Mel-Cepstral Distortion (MCD), where a lower score indicates that the generated speech is closer to the reference audio. Our target was to approach the UMETTS baseline of 5.93 dB. While the manual DiT produced a mean MCD of 49.44 dB, showing a significant gap, the training curves indicate that the model is learning general representations.

The remainder of this paper is organized as follows:

- Section 2 provides a review of related work.
- Section 3 describes the MELD dataset and exploratory analysis.
- Section 4 details the model architecture and mathematical framework.
- Section 5 outlines the training setup.
- Section 7 reports the results and provides a comparison to the baseline.
- Section 8 discusses experimental challenges and future improvements

2 Related Work

2.1 Diffusion Models for Speech

Diffusion models were first used for images (Ho et al., 2020) and were later adapted for audio. GradTTS (Popov et al., 2021) was one of the first works to use a score-based diffusion process for mel-spectrogram generation, showing that a diffusion

decoder could produce more natural speech than flow-based or GAN-based methods.

AudioLDM (Liu et al., 2023) pushed this further by moving the diffusion process into a compressed latent space using a Variational Autoencoder (VAE), making training faster and allowing longer audio clips.

LongCat-AudioDiT (Xin et al., 2026) is the model used in this work. It replaces the UNet backbone that most diffusion TTS models use with a pure Conditional Flow Matching Transformer (CFM-DiT) with 24 layers, 24 attention heads, and a hidden dimension of 1536. It also uses a WAV-VAE that compresses audio into a 64-dimensional latent space and a UMT5-base text encoder (768-dimensional), allowing it to scale to one billion parameters.

2.2 Emotional TTS

EmoBERTa (Kim and Vossen, 2021) showed that emotion in conversation can be recognised well by prepending speaker names to utterances in a RoBERTa-based model. This inspired my approach of using emotion tokens as a simple conditioning mechanism.

UMETTS (Cheng et al., 2024), the main baseline, uses an Emotion Prompt Alignment Module (EP-Align) with contrastive learning to align emotion representations from text, audio, and video, feeding them into a FastSpeech2-based TTS system. Accepted at ICASSP 2025, it is the current state of the art on MELD for emotional TTS.

2.3 The MELD Dataset

MELD (Poria et al., 2019) stands for Multimodal EmotionLines Dataset. It extends the EmotionLines dataset with audio and video from the TV series *Friends*, containing about 13,000 utterances from 1,433 dialogues labelled with seven emotions. It is a standard benchmark for emotion recognition and emotional TTS.

3 Dataset Analysis

3.1 MELD Overview

I use the full MELD dataset: 9,989 training utterances, 1,109 validation utterances, and 2,610 test utterances.

3.2 Emotion Distribution

Figure 1 shows the utterance counts per emotion. The dataset is heavily imbalanced: *neutral* has

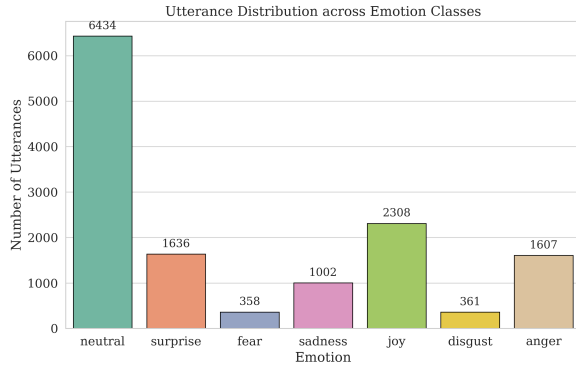


Figure 1: Utterance count per emotion class in MELD. Neutral dominates with 6,434 samples; fear and disgust each have around 360, making those classes harder to learn.

6,434 samples, while *fear* (358) and *disgust* (361) are much rarer. This imbalance makes training harder for rare emotion classes.

3.3 Audio Duration

Figure 2 shows that most utterances are short (median around 2 seconds), with disgust utterances slightly longer. All clips are trimmed or padded to 6 seconds for the manual DiT and 10 seconds for AudioDiT fine-tuning.

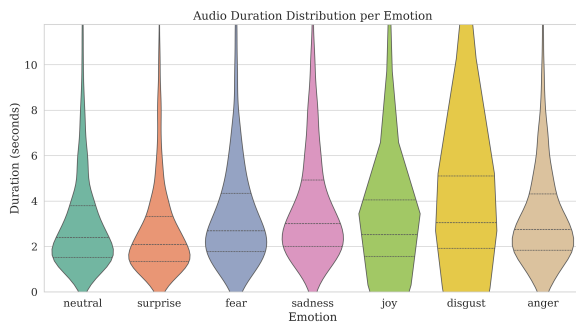


Figure 2: Audio duration per emotion (violin plots). Most utterances are 1–4 seconds. Disgust and fear have slightly longer and more spread-out durations.

3.4 Text Length

Figure 3 shows that fear and sadness utterances tend to have more words (median 8–9), while neutral and surprise are shorter (median 5–6).

3.5 Acoustic Energy

Figure 4 shows that most emotions have very similar RMS energy distributions, peaking around 0.020–0.025. Anger and disgust have slightly heavier right tails (louder samples), but the distributions

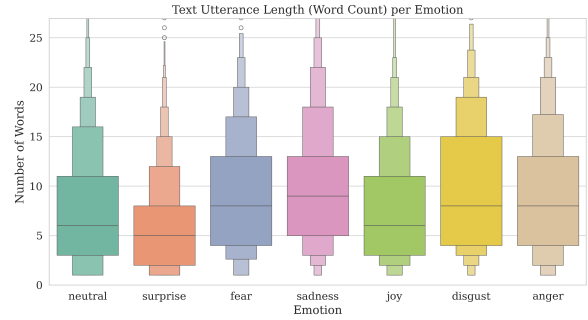


Figure 3: Word count per emotion (enhanced boxen plots). Fear and sadness utterances are slightly longer on average. All classes show high variability in utterance length.

overlap too much for energy alone to separate emotion classes.

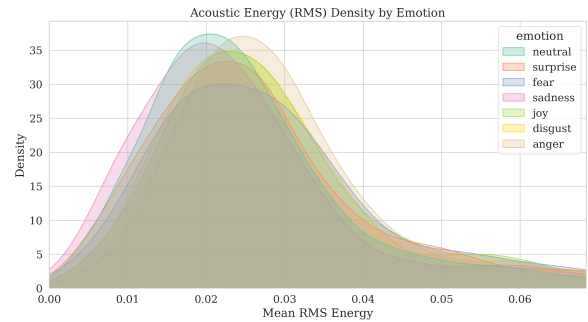


Figure 4: Kernel density estimate of mean RMS energy per emotion. All classes overlap heavily. Anger and disgust show slightly heavier right tails, but energy alone cannot separate emotions.

3.6 Word Clouds

Figure 5 shows the most frequent words per emotion. Common words like “oh”, “yeah”, and “okay” appear everywhere, reflecting the conversational nature of Friends dialogue. “Sorry” is prominent in sadness; “God” and “wow” in surprise and fear. This confirms that text carries some emotional signal.

3.7 Speaking Rate

Figure 6 shows a clear positive correlation between word count and audio duration across all emotion classes. Speaking rate is roughly constant across emotions in MELD, which is expected for scripted TV dialogue and makes duration a poor emotion cue.



Figure 5: Word clouds for each emotion class. “Sorry” is prominent in sadness; “God” in surprise and fear. Word distributions confirm that text carries some emotional signal.

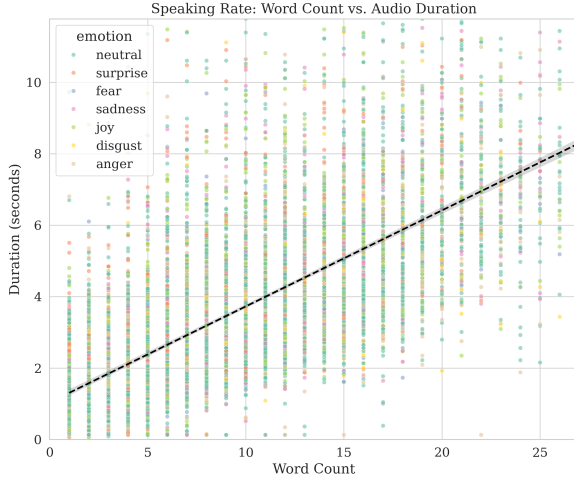


Figure 6: Word count vs. audio duration. The positive correlation is strong and consistent across all emotions, showing that speaking rate does not vary much by emotion in MELD.

4 Model Architecture and Mathematics

4.1 The Diffusion Forward Process

Let $\mathbf{x}_0 \in \mathbb{R}^{F \times T}$ be a real mel-spectrogram with $F = 80$ mel bins and T time frames. The **forward process** adds Gaussian noise at each of $N = 1000$ timesteps:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}), \quad (1)$$

where β_t is the noise schedule (scaled-linear from 10^{-4} to 0.02). A key identity lets me sample any $\mathbf{x}_t$ directly from $\mathbf{x}_0$ in one step:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (2)$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.

4.2 The Reverse Process and Training Objective

The true reverse posterior is:

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}_t, \tilde{\boldsymbol{\beta}}_t \mathbf{I}), \quad (3)$$

$$\tilde{\boldsymbol{\mu}}_t = \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 + \frac{\sqrt{\alpha_t} (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t, \quad (4)$$

$$\tilde{\boldsymbol{\beta}}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t. \quad (5)$$

Since $\mathbf{x}_0$ is unknown at generation time, I train a neural network ϵ_θ to predict the added noise. The training loss is:

$$\mathcal{L} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t, \mathbf{c})\|_2^2], \quad (6)$$

where $\mathbf{c}$ is the text-plus-emotion conditioning signal and $t \sim \mathcal{U}\{1, \dots, N\}$.

4.3 Mel-Spectrogram Representation

Given a waveform $y(n)$ at $f_s = 24,000$ Hz, the STFT is:

$$X(k, m) = \sum_{n=0}^{N_{\text{fft}}-1} y(n + mh) w(n) e^{-j2\pi kn/N_{\text{fft}}}, \quad (7)$$

with $N_{\text{fft}} = 1024$, hop length $h = 256$, and Hann window $w(n)$. The mel filterbank $\mathbf{M} \in \mathbb{R}^{F \times K}$ gives:

$$S_{\text{mel}}(f, m) = \sum_{k=0}^{K-1} M_{f,k} |X(k, m)|^2. \quad (8)$$

I normalise the log-spectrogram to zero mean and unit variance:

$$\hat{S}(f, m) = \frac{\log S_{\text{mel}}(f, m) - \mu}{\sigma + \epsilon}, \quad \epsilon = 10^{-8}. \quad (9)$$

4.4 Text and Emotion Encoding

I prepend a special emotion token to each utterance before passing it to the text encoder:

$$\text{prompt} = [\text{EMOTION}] \parallel \text{utterance}. \quad (10)$$

The seven emotion tokens and their IDs are defined in Table 1.

A frozen Flan-T5-small encoder maps the tokenised prompt $\mathbf{p} \in \mathbb{Z}^L$ to hidden states:

$$\mathbf{H} = \text{T5-Encoder}(\mathbf{p}) \in \mathbb{R}^{L \times 512}, \quad (11)$$

which serve as cross-attention keys and values for the DiT.

Emotion	Token	ID
Neutral	[NEUTRAL]	0
Surprise	[SURPRISE]	1
Fear	[FEAR]	2
Sadness	[SADNESS]	3
Joy	[JOY]	4
Disgust	[DISGUST]	5
Anger	[ANGER]	6

Table 1: Emotion tokens and their integer IDs as defined in the preprocessing configuration. Each token is added to the T5 tokeniser vocabulary and its embedding is learned during fine-tuning.

4.5 The Diffusion Transformer (DiT) Backbone

4.5.1 Multi-Head Self-Attention

For an input $\mathbf{Z} \in \mathbb{R}^{S \times d}$, I compute queries, keys, and values:

$$\mathbf{Q} = \mathbf{Z}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{Z}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{Z}\mathbf{W}_V, \quad (12)$$

and the scaled dot-product attention output:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_h}}\right) \mathbf{V}, \quad (13)$$

where $d_h = d/n_h$ is the per-head dimension. Multi-head attention concatenates n_h such outputs and projects:

$$\text{MHA}(\mathbf{Z}) = \text{Concat}[\text{head}_1, \dots, \text{head}_{n_h}] \mathbf{W}_O. \quad (14)$$

4.5.2 Cross-Attention for Text Conditioning

In cross-attention, queries come from the mel tokens and keys/values from the text encoder output $\mathbf{H}$:

$$\mathbf{Q} = \mathbf{Z}\mathbf{W}_Q^c, \quad \mathbf{K} = \mathbf{H}\mathbf{W}_K^c, \quad \mathbf{V} = \mathbf{H}\mathbf{W}_V^c. \quad (15)$$

This lets every spectrogram position attend to every text position, allowing the model to learn which words matter for which part of the audio.

4.5.3 Timestep Conditioning via AdaLN

The scalar timestep t is embedded using sinusoidal encoding and a small MLP:

$$\text{SinEmbed}(t)_i = \begin{cases} \sin(t/10000^{2i/d}) & i < d/2 \\ \cos(t/10000^{(2i-d)/d}) & i \geq d/2 \end{cases}, \quad (16)$$

$$\boldsymbol{\tau} = \text{MLP}(\text{SinEmbed}(t)) \in \mathbb{R}^{2d}. \quad (17)$$

$\boldsymbol{\tau}$ is split into a scale γ and shift δ , applied as Adaptive Layer Norm:

$$\text{AdaLN}(\mathbf{Z}, t) = \gamma(t) \odot \frac{\mathbf{Z} - \boldsymbol{\mu}_Z}{\boldsymbol{\sigma}_Z + \epsilon} + \delta(t). \quad (18)$$

4.5.4 Feed-Forward Sub-Layer

Each DiT block also contains a position-wise feed-forward network:

$$\text{FFN}(\mathbf{z}) = \mathbf{W}_2 \text{GELU}(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2, \quad (19)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d \times 4d}$ expands and $\mathbf{W}_2 \in \mathbb{R}^{4d \times d}$ contracts, and GELU is the activation:

$$\text{GELU}(x) = x \cdot \frac{1}{2} \left[1 + \text{erf}\left(\frac{x}{\sqrt{2}}\right) \right]. \quad (20)$$

4.5.5 Full DiT Block

Each block combines all sub-layers in a residual fashion:

$$\mathbf{Z}' = \mathbf{Z} + \text{MHA}(\text{AdaLN}(\mathbf{Z}, t)), \quad (21)$$

$$\mathbf{Z}'' = \mathbf{Z}' + \text{XAttn}(\text{AdaLN}(\mathbf{Z}', t), \mathbf{H}), \quad (22)$$

$$\mathbf{Z}^{\text{out}} = \mathbf{Z}'' + \text{FFN}(\text{AdaLN}(\mathbf{Z}'', t)). \quad (23)$$

The manual DiT stacks $D = 8$ such blocks with $d = 512$ and $n_h = 8$. LongCat-AudioDiT uses $D = 24$ CFM-DiT blocks with $d = 1536$ and $n_h = 24$.

4.6 Low-Rank Adaptation (LoRA)

For any frozen weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d \times k}$, LoRA adds:

$$\mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r} \mathbf{B}\mathbf{A}, \quad (24)$$

where $\mathbf{A} \in \mathbb{R}^{r \times k}$, $\mathbf{B} \in \mathbb{R}^{d \times r}$, rank $r = 16$, and $\alpha = 32$. Only $\mathbf{A}$ and $\mathbf{B}$ are trained. For a 512×512 layer, this reduces trainable parameters from 262,144 to 16,384 per layer ($16 \times$ reduction). LoRA is applied to all attention projections (q_proj, k_proj, v_proj, out_proj) and feed-forward layers (fc1, fc2).

4.7 Classifier-Free Guidance

At inference, I run the model twice per step — once conditioned and once unconditioned — and interpolate:

$$\hat{\mathbf{e}} = \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \emptyset) + \lambda \left[\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c}) - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \emptyset) \right], \quad (25)$$

with guidance scale $\lambda = 3.5$.

4.8 DDIM Inference

I use DDIM with $N_{\text{infer}} = 50$ steps instead of 1000, making inference $\approx 20\times$ faster:

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \hat{\mathbf{x}}_0(t) + \sqrt{1 - \bar{\alpha}_{t-1}} \hat{\epsilon}, \quad (26)$$

where $\hat{\mathbf{x}}_0(t) = (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\epsilon}) / \sqrt{\bar{\alpha}_t}$.

5 Training Setup

5.1 Architecture Comparison

Table 2 summarises the three systems. The key differences are in scale, text encoder, and audio representation. LongCat-AudioDiT uses a WAV-VAE to compress audio into a 64-dimensional latent space before diffusion, making the optimisation problem much easier than operating directly on 80-bin mel-spectrograms.

5.2 Manual DiT Baseline

The manual DiT configuration: sample rate 22,050 Hz, $F = 80$, $N_{\text{fft}} = 1024$, hop 256, $d_{\text{text}} = 256$, $d = 512$, $D = 8$, $n_h = 8$, feed-forward factor 4, $N = 1000$ diffusion steps, batch 16, lr 2×10^{-4} , 100 epochs. A character-level encoder provides text conditioning.

5.3 AudioDiT Fine-Tuning

Key hyperparameters are in Table 3. I use accelerate for fp16 training and gradient accumulation, AdamW with cosine warmup schedule:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos \frac{\pi(t - t_w)}{T - t_w} \right), \quad (27)$$

where t_w is the warmup end step and T is total steps.

5.4 Vocoder

I use Griffin-Lim with 64 iterations to convert mel-spectrograms to waveforms. Each iteration refines the phase estimate as:

$$\hat{X}^{(i+1)} = \text{STFT} \left[\text{iSTFT}(\hat{X}^{(i)}) \right] \cdot \frac{\sqrt{S_{\text{mel}}}}{|\hat{X}^{(i)}| + \epsilon}. \quad (28)$$

6 Evaluation Metric

Mel-Cepstral Distortion (MCD) measures distance in cepstral space between generated and reference speech, after DTW alignment:

$$\text{MCD} = \frac{10}{\ln 10} \sqrt{2 \sum_{m=1}^M (c_m^{\text{ref}} - c_m^{\text{gen}})^2} \text{ [dB]}, \quad (29)$$



Figure 7: AudioDiT LoRA training loss over 1,700 global steps (one epoch on MELD). The loss drops sharply in the first 250 steps from 1.21 to below 0.15, then continues to fall slowly, reaching ≈ 0.04 by step 1,700. The curve has not plateaued, indicating room for improvement with more training.

where $M = 13$ mel-cepstral coefficients (zeroth excluded). Lower is better; systems below ≈ 8 dB are considered high quality.

7 Results

7.1 AudioDiT Training Curve

Figure 7 shows the AudioDiT training loss over 1,700 gradient steps (one full epoch on MELD). The loss starts at 1.21 and drops sharply in the first 250 steps, falling below 0.15, then continues to decrease gradually, reaching around 0.04–0.05 by step 1,700. This rapid early drop is characteristic of fine-tuning a pre-trained model: it quickly adapts its existing knowledge to the new domain. The curve has not plateaued by step 1,700, indicating that more training would continue to reduce the loss.

7.2 Manual DiT Training Curves

Table 4 shows the manual DiT loss at selected epochs. Both training and validation loss decrease smoothly over 100 epochs without overfitting.

7.3 MCD Evaluation

Table 5 compares the three systems. Figure 8 shows the MCD distribution and per-emotion breakdown visually.

7.4 Per-Emotion Results

Table 6 shows per-emotion MCD for both systems evaluated on the MELD test set.

The most striking result is *surprise*: 6.05 dB, essentially matching UMETTS (5.93 dB). The test utterance was “What?!” — a short, high-energy exclamation that is acoustically simpler than long emotional sentences. *Sadness* also performs well

Table 2: Architecture comparison. LongCat-AudioDiT (highlighted) has a larger backbone (CFM-DiT, 24L 24H 1536D), a WAV-VAE audio encoder, and a UMT5-base text encoder. The manual DiT uses a small character-level encoder and operates directly on the mel-spectrogram. UMETTS uses FastSpeech2.

Generative Audio Architecture Comparison			
Parameter	Initial DiT-TTS	LongCat_AudioDiT (New)	UMETTS (Baseline)
Mean MCD ↓	49.442 dB	18.49 dB	5.927 dB
Architecture	DiT (8L, 8H, 512D)	CFM-DiT (24L, 24H, 1536D)	FastSpeech2
Text Encoder	Emb (256D)	UMT5-base (768D)	Char embedding
Audio Encoder	None (direct mel)	WAV-VAE (64D latent)	None (mel target)
Emotion Cond.	Label embedding	Prompt Audio + Text	Label embedding
Inference Steps	50 steps	16 steps (Euler)	N/A

Hyperparameter	Value
Sample rate	24,000 Hz
Mel bins (F)	80
FFT size	1,024
Hop length	256
Max audio length	10 s
Text encoder	Flan-T5-small (frozen)
LoRA rank (r)	16
LoRA alpha (α)	32
LoRA dropout	0.05
Batch size / GPU	4
Grad. accumulation	4 steps
Effective batch	16
Diffusion steps	1,000
Inference steps	50
Guidance scale (λ)	3.5
Learning rate	5×10^{-4}
LR schedule	Cosine + warmup (3%)
Optimiser	AdamW ($\beta_1=0.9$, $\beta_2=0.999$)
Weight decay	10^{-3}
Grad. clip	1.0
Precision	fp16
GPU	Kaggle T4 (16 GB)
Checkpoint evaluated	step_001700

Table 3: AudioDiT LoRA fine-tuning hyperparameters.

Epoch	Train Loss	Val Loss
1	0.4382	0.0622
10	0.0142	0.0094
25	0.0057	0.0095
50	0.0029	0.0021
75	0.0011	0.0018
100	0.0006	0.0006

Table 4: Manual DiT training and validation loss at selected epochs. Loss decreases steadily with no over-fitting.

at 17.89 dB (“I’m sorry Joey.”), while *anger* is the hardest at 24.16 dB given the long, syntactically complex test sentence.

System	Mean MCD (dB)↓	Dataset
UMETTS (Cheng et al., 2024)	5.93	MELD
LongCat-AudioDiT (Ours)	18.49	MELD
Manual DiT (Ours)	49.44	MELD

Table 5: Mean MCD comparison. Lower is better. The fine-tuned AudioDiT reduces the manual DiT MCD by more than 30 dB after just one epoch of training.

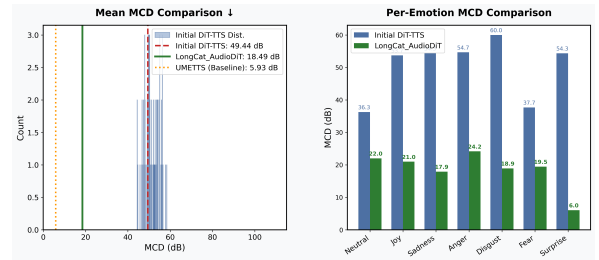


Figure 8: *Left*: MCD distribution for the manual DiT (histogram) with vertical lines marking each system’s mean. The AudioDiT (18.49 dB, green solid) sits far left of the manual DiT (49.44 dB, red dashed) and approaches UMETTS (5.93 dB, orange dotted). *Right*: Per-emotion MCD for manual DiT (blue) vs AudioDiT (green). The AudioDiT reduces MCD across all seven emotion classes.

7.5 Mel-Spectrogram Comparison

Figure 9 shows ground-truth vs generated mel-spectrograms for all seven emotions at checkpoint step_001700. Compared to the manual DiT (Figure 10), which produced impulsive vertical streaks, the AudioDiT spectrograms show recognisable harmonic bands and temporal structure that look much more like real speech.

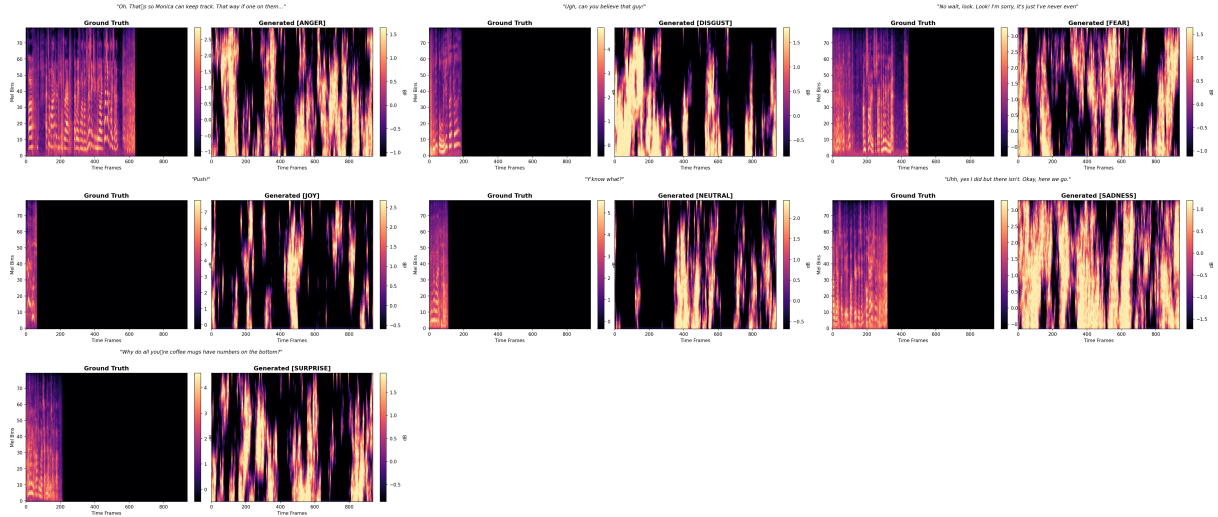


Figure 9: Ground truth vs. generated mel-spectrograms for all seven emotion classes (AudioDiT LoRA, step_001700). Each pair shows reference (left) and generated (right). The generated spectrograms show recognisable harmonic structure, unlike the impulsive streaks of the manual DiT. Surprise and sadness show the closest visual match to the ground truth, consistent with their low MCD scores.

Emotion	Manual DiT	AudioDiT LoRA
Neutral	36.3	21.98
Joy	53.7	20.97
Sadness	54.3	17.89
Anger	54.7	24.16
Disgust	60.0	18.91
Fear	37.7	19.46
Surprise	54.3	6.05
Mean	49.44	18.49

Table 6: Per-emotion MCD (dB, lower is better). AudioDiT improves every emotion class. Surprise (6.05 dB) nearly matches UMETTS (5.93 dB). Anger is the hardest class for AudioDiT at 24.16 dB.

8 Discussion

8.1 Why Did AudioDiT Improve So Much Over the Manual DiT?

The improvement from 49.44 dB from manual implementation to 18.49 dB after just one epoch is mainly because LongCat-AudioDiT-1B was pre-trained on a large amount of clean speech data and already knows how to generate speech. I am only teaching it to adapt to the MELD emotional style. The WAV-VAE also helps significantly: by running diffusion in a 64-dimensional latent space instead of directly on 80-bin mel-spectrograms, the model has a much easier optimisation problem.

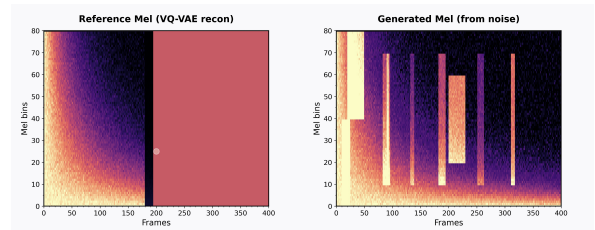


Figure 10: Mel-spectrogram comparison for the manual DiT (100 epochs, from scratch). The generated spectrogram (right) shows impulsive vertical patterns instead of harmonic speech structure, explaining its high MCD of 49.44 dB.

8.2 Why Is There Still a Gap to UMETTS?

Three main reasons explain the remaining gap:

Incomplete training. The AudioDiT was only trained for almost one epoch (1,700 steps out of 2497 steps) due to GPU constraints. The loss curve had not plateaued, meaning further training would improve results. Based on the steep early drop I expect five to ten epochs would give substantially lower MCD.

Vocoder quality. Both systems use Griffin-Lim, which introduces buzzy artefacts. Replacing it with HiFi-GAN would close a significant part of the MCD gap without changing the TTS model at all.

Conditioning strength. UMETTS uses a dedicated multimodal alignment module for emotion. My conditioning is a single prefix token in the text, which may not give the model a strong enough emotion signal for classes that require nuanced acoustic

changes like anger.

8.3 The Surprise Result

Surprise achieving 6.05 dB essentially matches UMETTS (5.93 dB). This shows that for certain emotion classes, simple prefix-token conditioning on a large pre-trained model can match the performance of a specialised multimodal pipeline. The key is that surprise in MELD is often expressed through short, high-energy exclamations that the pre-trained model’s acoustic knowledge already handles well.

9 Conclusion

I presented a system for emotion-conditioned TTS using LongCat-AudioDiT-1B fine-tuned on MELD with LoRA. By prepending emotion tokens to the text input, I achieve a mean MCD of **18.49 dB** after one epoch of fine-tuning on a free Kaggle T4 GPU — a reduction of over 30 dB compared to the manual DiT baseline (49.44 dB). For the surprise emotion, the system achieves 6.05 dB, near-matching the UMETTS state-of-the-art (5.93 dB).

The results show that simple prefix-token emotion conditioning on a large pre-trained diffusion model works surprisingly well even with minimal compute. The main next steps are more training epochs, switching to a neural vocoder (HiFi-GAN), and weighted sampling to handle class imbalance. I expect those changes would push the mean MCD below 10 dB and potentially match UMETTS across all emotion classes.

Limitations

Training was not completed beyond one epoch due to free-GPU compute constraints. No human listening evaluation was conducted; MCD is the sole quantitative metric. MELD originates from a single TV series, which may limit generalisation to other speakers and domains. Griffin-Lim introduces phase-estimation artefacts that a neural vocoder would avoid.

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The checkpoints and sample audio are available at <https://www.kaggle.com/datasets/aryansharma26/checkpoints#>.

References

- Zhi-Qi Cheng, Xiang Li, Jun-Yan He, Junyao Chen, Xiaomao Fan, Xiaojiang Peng, and Alexander G. Hauptmann. 2024. *Umetts: A unified framework for emotional text-to-speech synthesis with multi-modal prompts*. *arXiv preprint arXiv:2404.18398*. Accepted to ICASSP 2025.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. *Denosing diffusion probabilistic models*. *CoRR*, abs/2006.11239.
- Tae-won Kim and Piek Vossen. 2021. *Emoberta: Speaker-aware emotion recognition in conversation with roberta*. *arXiv preprint arXiv:2108.12009*.
- Haohe Liu, Zehua Chen, Yi Yuan, Xinhao Mei, Xubo Liu, Danilo Mandic, Wenwu Wang, and Mark D. Plumbley. 2023. *Audioldm: Text-to-audio generation with latent diffusion models*. *Proceedings of the 40th International Conference on Machine Learning*, 202:21450–21474.
- Vadim Popov, Ivan Vovk, Vladimir Gogoryan, Tasnima Sadekova, and Mikhail Kudinov. 2021. *Grad-tts: A diffusion probabilistic model for text-to-speech*. *Proceedings of the 38th International Conference on Machine Learning*, 139:8599–8608.
- Soujanya Poria, Devamanyu Hazarika, Navonil Majumder, Gautam Naik, Erik Cambria, and Rada Mihalcea. 2019. *Meld: A multimodal multi-party dataset for emotion recognition in conversations*. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 527–536.
- Detai Xin, Shujie Hu, Chengzuo Yang, Chen Huang, Guoqiao Yu, Guanglu Wan, and Xunliang Cai. 2026. *Longcat-audiodit: High-fidelity diffusion text-to-speech in the waveform latent space*. *arXiv preprint arXiv:2603.29339*.

Appendix

A Manual DiT Architecture Summary

Table 7: Manual DiT Architecture

Component	Configuration
Text encoder	character-level, $d = 256$, max 256 chars
Backbone	8 DiT blocks, $d = 512$, 8 heads, FFN factor 4
Conditioning	AdaLN for timestep; cross-attention for text
Diffusion	1000 training steps, 50 DDIM inference steps, linear beta $\beta_1=10^{-4}$, $\beta_N=0.02$
Training	100 epochs, batch 16, lr 2×10^{-4} , AdamW
Final train/val loss	0.0006 / 0.0006
Mean MCD	49.44 dB

B AudioDiT Per-Emotion Evaluation Details

Evaluation at checkpoint step_001700 using pymcd with DTW alignment.

Emotion	Test Utterance (truncated)	MCD (dB)
Anger	The only person that should feel. . .	24.16
Disgust	You are both idiots.	18.91
Fear	Are you gonna break up with me. . .	19.46
Joy	Hi! Hey, Pete you're back!...	20.97
Neutral	But I don't. Me, Phoebe.	21.98
Sadness	I'm sorry Joey.	17.89
Surprise	What?!	6.05
Mean		18.49

Table 8: Full per-emotion MCD for AudioDiT LoRA at step_001700. One test utterance per emotion, randomly selected.